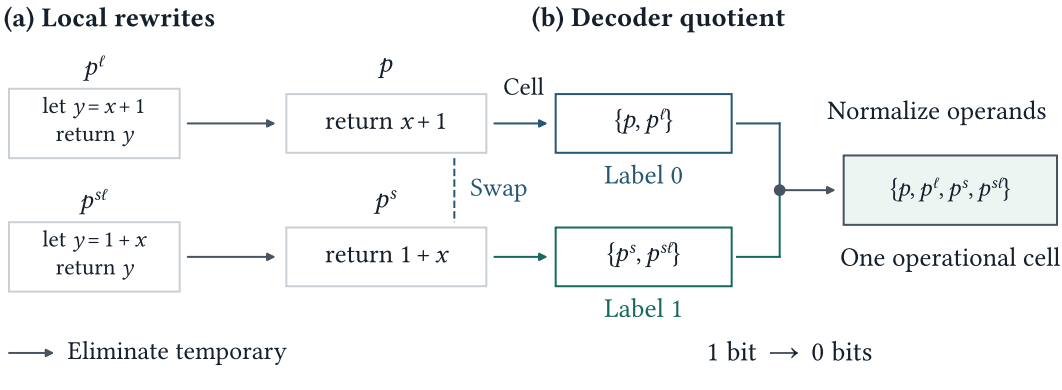
(a) Local rewrites
(b) Decoder quotient
p^{ℓ}
let $y = x + 1$
return y
p
return $x + 1$
Cell
$\{p, p^{\ell}\}$
Label 0
Normalize operands
$p^{s\ell}$
let $y = 1 + x$
return y
p^{s}
return $1 + x$
Swap
$\{p^{s}, p^{s\ell}\}$
Label 1
$\{p, p^{\ell}, p^{s}, p^{s\ell}\}$
One operational cell
Eliminate temporary
1 bit ⟶ 0 bits